

## 8 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY

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### KEY FINDINGS

- The sequencing by synthesis (SBS) segment accounted for the largest share of 45% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 253.56 million by 2028 from USD 129.01 million in 2023, at a CAGR of 14.5%.
- The nanopore sequencing segment is estimated to grow at the highest CAGR of 17.3% from 2023 to 2028. This segment is projected to reach USD 34.21 million by 2028.
- Other sequencing technologies (Sanger sequencing and semiconductor sequencing) form the second-largest segment in this market, with a share of ~37% in 2022.

## 8.1 INTRODUCTION

Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share (~45%) in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR (17.3%) during the forecast period.

**TABLE 50** MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

| Technology              | 2021          | 2022          | 2023          | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| Sequencing by Synthesis | 99.59         | 113.38        | 129.01        | 146.93        | 167.48        | 191.59        | 219.92        | 253.56        | 14.5%               |
| Sequencing by Ligation  | 28.18         | 31.96         | 36.24         | 41.13         | 46.71         | 53.24         | 60.89         | 69.95         | 14.1%               |
| Nanopore Sequencing     | 11.29         | 13.20         | 15.42         | 18.01         | 21.05         | 24.67         | 29.00         | 34.21         | 17.3%               |
| Other Technologies      | 80.97         | 91.63         | 103.64        | 117.32        | 132.92        | 151.11        | 172.40        | 197.53        | 13.8%               |
| <b>Total</b>            | <b>220.03</b> | <b>250.17</b> | <b>284.31</b> | <b>323.39</b> | <b>368.16</b> | <b>420.61</b> | <b>482.21</b> | <b>555.24</b> | <b>14.3%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 8.2 SEQUENCING BY SYNTHESIS (SBS)

### 8.2.1 SBS TECHNOLOGY ENABLES HIGHEST PRODUCTION OF BASE PAIRS IN SHORT TIME

The SBS technology uses four fluorescently labeled nucleotides to sequence millions of clusters on the flow cell surface in parallel. During every sequencing cycle, a single-labeled deoxyribonucleoside triphosphate (dNTP) is added to the nucleic acid chain. This nucleotide label serves as a terminator for polymerization. After dNTP incorporation, the dye is first imaged to identify the base, and then it is enzymatically cleaved to allow the incorporation of another nucleotide. Since all four reversible terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias. The result is true base-by-base sequencing that enables the most accurate data for a broad range of applications.

The SBS technology has a wide customer base owing to its well-established precision in sequencing as well as its broad application base. It has applications in microbial genetics and metagenomics and is mainly used in whole-genome sequencing, transcriptome analysis, targeted sequencing, and genome-wide protein-nucleic acid interaction analysis.

**TABLE 51** MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022          | 2023          | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|------------------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| North America          | 57.03        | 64.63         | 73.20         | 82.98         | 94.15         | 107.20        | 122.48        | 140.55        | 13.9%               |
| Europe                 | 30.83        | 35.22         | 40.20         | 45.94         | 52.54         | 60.30         | 69.44         | 80.32         | 14.8%               |
| Asia Pacific           | 8.47         | 9.93          | 11.63         | 13.62         | 15.96         | 18.75         | 22.09         | 26.12         | 17.6%               |
| Latin America          | 1.93         | 2.14          | 2.37          | 2.63          | 2.92          | 3.25          | 3.62          | 4.05          | 11.3%               |
| Middle East and Africa | 1.33         | 1.46          | 1.60          | 1.75          | 1.91          | 2.10          | 2.30          | 2.52          | 9.5%                |
| <b>Total</b>           | <b>99.59</b> | <b>113.38</b> | <b>129.01</b> | <b>146.93</b> | <b>167.48</b> | <b>191.59</b> | <b>219.92</b> | <b>253.56</b> | <b>14.5%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 52** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021         | 2022         | 2023         | 2024         | 2025         | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------------|
| US           | 51.68        | 58.62        | 66.45        | 75.39        | 85.60        | 97.54         | 111.54        | 128.09        | 14.0%               |
| Canada       | 5.34         | 6.01         | 6.75         | 7.59         | 8.55         | 9.66          | 10.94         | 12.46         | 13.0%               |
| <b>Total</b> | <b>57.03</b> | <b>64.63</b> | <b>73.20</b> | <b>82.98</b> | <b>94.15</b> | <b>107.20</b> | <b>122.48</b> | <b>140.55</b> | <b>13.9%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 53** EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Germany        | 6.68         | 7.65         | 8.77         | 10.04        | 11.52        | 13.26        | 15.32        | 17.77        | 15.2%               |
| UK             | 5.97         | 6.87         | 7.91         | 9.10         | 10.48        | 12.12        | 14.06        | 16.38        | 15.7%               |
| France         | 4.72         | 5.39         | 6.15         | 7.03         | 8.05         | 9.24         | 10.64        | 12.31        | 14.9%               |
| Italy          | 1.33         | 1.51         | 1.72         | 1.96         | 2.24         | 2.56         | 2.94         | 3.39         | 14.5%               |
| Spain          | 1.02         | 1.16         | 1.32         | 1.50         | 1.71         | 1.95         | 2.24         | 2.58         | 14.3%               |
| Rest of Europe | 11.11        | 12.62        | 14.34        | 16.30        | 18.54        | 21.16        | 24.24        | 27.89        | 14.2%               |
| <b>Total</b>   | <b>30.83</b> | <b>35.22</b> | <b>40.20</b> | <b>45.94</b> | <b>52.54</b> | <b>60.30</b> | <b>69.44</b> | <b>80.32</b> | <b>14.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 54** ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region       | 2021        | 2022        | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| China                | 3.58        | 4.21        | 4.93         | 5.77         | 6.77         | 7.96         | 9.38         | 11.09        | 17.6%               |
| Japan                | 1.67        | 1.97        | 2.32         | 2.73         | 3.21         | 3.80         | 4.50         | 5.34         | 18.2%               |
| India                | 0.21        | 0.25        | 0.29         | 0.34         | 0.40         | 0.46         | 0.55         | 0.64         | 17.3%               |
| Rest of Asia Pacific | 3.01        | 3.51        | 4.10         | 4.78         | 5.58         | 6.53         | 7.67         | 9.03         | 17.1%               |
| <b>Total</b>         | <b>8.47</b> | <b>9.93</b> | <b>11.63</b> | <b>13.62</b> | <b>15.96</b> | <b>18.75</b> | <b>22.09</b> | <b>26.12</b> | <b>17.6%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 55** LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region        | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Brazil                | 0.91        | 1.01        | 1.12        | 1.24        | 1.38        | 1.53        | 1.71        | 1.92        | 11.4%               |
| Rest of Latin America | 1.02        | 1.13        | 1.26        | 1.39        | 1.54        | 1.71        | 1.91        | 2.13        | 11.2%               |
| <b>Total</b>          | <b>1.93</b> | <b>2.14</b> | <b>2.37</b> | <b>2.63</b> | <b>2.92</b> | <b>3.25</b> | <b>3.62</b> | <b>4.05</b> | <b>11.3%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 8.3 SEQUENCING BY LIGATION (SBL)

### 8.3.1 INCREASING DEMAND FOR SBL IN FOOD MICROBIOME RESEARCH TO BOOST GROWTH

SBL is a DNA sequencing method that uses DNA ligase to identify the nucleotide present at a given position in a DNA sequence. SBL analyzes the mismatch in base pairs of a DNA ligase to determine the underlying sequence of a target DNA molecule.

SOLiD is an enzymatic method of sequencing that uses DNA ligase, an enzyme used widely in biotechnology, for its ability to ligate double-stranded DNA strands. Emulsion PCR is used to immobilize/amplify an ssDNA primer-binding region (known as an adapter), which has been conjugated to the target sequence on a bead. These beads are then deposited onto a glass surface with a high density of beads. This, in turn, increases the throughput of the technique.

**TABLE 56** MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| North America          | 16.07        | 18.15        | 20.49        | 23.15        | 26.18        | 29.71        | 33.83        | 38.69        | 13.6%               |
| Europe                 | 8.75         | 9.96         | 11.32        | 12.89        | 14.69        | 16.80        | 19.27        | 22.21        | 14.4%               |
| Asia Pacific           | 2.42         | 2.83         | 3.29         | 3.84         | 4.48         | 5.23         | 6.13         | 7.22         | 17.0%               |
| Latin America          | 0.55         | 0.61         | 0.68         | 0.75         | 0.82         | 0.91         | 1.01         | 1.13         | 10.8%               |
| Middle East and Africa | 0.38         | 0.42         | 0.46         | 0.50         | 0.54         | 0.59         | 0.65         | 0.71         | 9.2%                |
| <b>Total</b>           | <b>28.18</b> | <b>31.96</b> | <b>36.24</b> | <b>41.13</b> | <b>46.71</b> | <b>53.24</b> | <b>60.89</b> | <b>69.95</b> | <b>14.1%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 57** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| US           | 14.56        | 16.46        | 18.60        | 21.03        | 23.79        | 27.02        | 30.79        | 35.23        | 13.6%               |
| Canada       | 1.51         | 1.69         | 1.89         | 2.13         | 2.39         | 2.69         | 3.04         | 3.45         | 12.8%               |
| <b>Total</b> | <b>16.07</b> | <b>18.15</b> | <b>20.49</b> | <b>23.15</b> | <b>26.18</b> | <b>29.71</b> | <b>33.83</b> | <b>38.69</b> | <b>13.6%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 58** EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region | 2021        | 2022        | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|----------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| Germany        | 1.89        | 2.16        | 2.46         | 2.80         | 3.20         | 3.67         | 4.22         | 4.87         | 14.7%               |
| UK             | 1.69        | 1.94        | 2.22         | 2.55         | 2.93         | 3.37         | 3.90         | 4.52         | 15.3%               |
| France         | 1.34        | 1.52        | 1.73         | 1.98         | 2.26         | 2.58         | 2.97         | 3.43         | 14.6%               |
| Italy          | 0.38        | 0.43        | 0.49         | 0.55         | 0.63         | 0.72         | 0.83         | 0.95         | 14.3%               |
| Spain          | 0.29        | 0.33        | 0.37         | 0.42         | 0.48         | 0.55         | 0.62         | 0.72         | 14.0%               |
| Rest of Europe | 3.16        | 3.58        | 4.05         | 4.59         | 5.19         | 5.91         | 6.74         | 7.72         | 13.8%               |
| <b>Total</b>   | <b>8.75</b> | <b>9.96</b> | <b>11.32</b> | <b>12.89</b> | <b>14.69</b> | <b>16.80</b> | <b>19.27</b> | <b>22.21</b> | <b>14.4%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 59 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)**

| Country/Region       | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| China                | 1.02        | 1.19        | 1.39        | 1.62        | 1.89        | 2.21        | 2.58        | 3.04        | 16.9%               |
| Japan                | 0.48        | 0.56        | 0.66        | 0.77        | 0.90        | 1.06        | 1.25        | 1.47        | 17.6%               |
| India                | 0.06        | 0.07        | 0.08        | 0.10        | 0.11        | 0.13        | 0.15        | 0.18        | 16.8%               |
| Rest of Asia Pacific | 0.86        | 1.00        | 1.17        | 1.36        | 1.58        | 1.84        | 2.15        | 2.53        | 16.7%               |
| <b>Total</b>         | <b>2.42</b> | <b>2.83</b> | <b>3.29</b> | <b>3.84</b> | <b>4.48</b> | <b>5.23</b> | <b>6.13</b> | <b>7.22</b> | <b>17.0%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 60 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)**

| Country/Region        | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Brazil                | 0.26        | 0.29        | 0.32        | 0.35        | 0.39        | 0.43        | 0.48        | 0.53        | 10.8%               |
| Rest of Latin America | 0.29        | 0.32        | 0.36        | 0.39        | 0.44        | 0.48        | 0.53        | 0.60        | 10.7%               |
| <b>Total</b>          | <b>0.55</b> | <b>0.61</b> | <b>0.68</b> | <b>0.75</b> | <b>0.82</b> | <b>0.91</b> | <b>1.01</b> | <b>1.13</b> | <b>10.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

## 8.4 NANOPORE SEQUENCING

### 8.4.1 ONLY SEQUENCING TECHNOLOGY WITH DIRECT RNA SEQUENCING CAPABILITIES

Introduced by Oxford Nanopore Technologies, nanopore sequencing technology (which is considered the fourth generation of sequencing) has emerged as a competitive, portable technology.

The technology works through the detection of unique electrical signals of different molecules as they pass across the nanopore with a semiconductor-based electronic detection system. This technology makes for a high throughput, cost-effective sequencing solution. The technology's core/heart consists of the biological nanopore, a protein pore embedded in a membrane, while the brains of the technology lie in the electronics of a semiconductor integrated circuit and proprietary chemistries. The electronic sensor embedded in the chip enables automatic membrane assembly and nanopore insertion while allowing for active control of individual sensors on the circuit. Various types of sequencing chemistries can be paired with nanopore and electronic sensor technology, which enables high-throughput and high-accuracy sequencing. However, the low accuracy of nanopore sequencing as compared to other well-established techniques is one of the major factors restraining the growth of this market segment.

In mid-2014, Oxford Nanopore Technologies launched MinION, which was the only nanopore sequencer commercially available in the market until 2015. This sequencer is portable, weighs just 100 gm, and can sequence DNA and RNA of up to ~200kb in length. Due to its portable nature, the MinION sequencer can

be used for real-time sequencing. From January to July 2019, the company launched GridION, Flongle sequencer, and PromethION 24 and PromethION 48 (on the e-Commerce platform of the company to make them available globally). Also, in May 2018, the company upgraded the MinKnow software that drives the MinION, GridION, and PromethION sequencing devices.

**TABLE 61** MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------------|
| North America          | 6.47         | 7.47         | 8.62         | 9.95         | 11.49        | 13.30        | 15.46        | 18.03        | 15.9%               |
| Europe                 | 3.49         | 4.12         | 4.86         | 5.72         | 6.74         | 7.96         | 9.43         | 11.21        | 18.2%               |
| Asia Pacific           | 0.96         | 1.18         | 1.44         | 1.76         | 2.14         | 2.62         | 3.20         | 3.92         | 22.2%               |
| Latin America          | 0.22         | 0.26         | 0.30         | 0.35         | 0.41         | 0.47         | 0.55         | 0.64         | 16.5%               |
| Middle East and Africa | 0.15         | 0.18         | 0.20         | 0.24         | 0.27         | 0.31         | 0.36         | 0.41         | 15.2%               |
| <b>Total</b>           | <b>11.29</b> | <b>13.20</b> | <b>15.42</b> | <b>18.01</b> | <b>21.05</b> | <b>24.67</b> | <b>29.00</b> | <b>34.21</b> | <b>17.3%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 62** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021        | 2022        | 2023        | 2024        | 2025         | 2026         | 2027         | 2028         | CAGR<br>(2023–2028) |
|--------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|---------------------|
| US           | 5.87        | 6.78        | 7.82        | 9.03        | 10.43        | 12.09        | 14.05        | 16.40        | 16.0%               |
| Canada       | 0.61        | 0.70        | 0.80        | 0.92        | 1.06         | 1.22         | 1.41         | 1.63         | 15.4%               |
| <b>Total</b> | <b>6.47</b> | <b>7.47</b> | <b>8.62</b> | <b>9.95</b> | <b>11.49</b> | <b>13.30</b> | <b>15.46</b> | <b>18.03</b> | <b>15.9%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis



**TABLE 63** EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028         | CAGR<br>(2023–2028) |
|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|---------------------|
| Germany        | 0.76        | 0.89        | 1.04        | 1.23        | 1.44        | 1.70        | 2.00        | 2.38         | 17.9%               |
| UK             | 0.68        | 0.80        | 0.95        | 1.12        | 1.33        | 1.58        | 1.88        | 2.24         | 18.8%               |
| France         | 0.53        | 0.63        | 0.74        | 0.88        | 1.03        | 1.22        | 1.45        | 1.72         | 18.3%               |
| Italy          | 0.15        | 0.18        | 0.21        | 0.25        | 0.29        | 0.34        | 0.40        | 0.48         | 18.1%               |
| Spain          | 0.12        | 0.14        | 0.16        | 0.19        | 0.22        | 0.26        | 0.31        | 0.37         | 18.0%               |
| Rest of Europe | 1.26        | 1.49        | 1.75        | 2.06        | 2.43        | 2.86        | 3.39        | 4.02         | 18.1%               |
| <b>Total</b>   | <b>3.49</b> | <b>4.12</b> | <b>4.86</b> | <b>5.72</b> | <b>6.74</b> | <b>7.96</b> | <b>9.43</b> | <b>11.21</b> | <b>18.2%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 64** ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region       | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| China                | 0.41        | 0.50        | 0.61        | 0.74        | 0.90        | 1.09        | 1.33        | 1.63        | 21.9%               |
| Japan                | 0.19        | 0.23        | 0.29        | 0.35        | 0.43        | 0.53        | 0.65        | 0.80        | 22.8%               |
| India                | 0.02        | 0.03        | 0.04        | 0.04        | 0.05        | 0.07        | 0.08        | 0.10        | 22.2%               |
| Rest of Asia Pacific | 0.34        | 0.42        | 0.51        | 0.63        | 0.76        | 0.93        | 1.14        | 1.40        | 22.2%               |
| <b>Total</b>         | <b>0.96</b> | <b>1.18</b> | <b>1.44</b> | <b>1.76</b> | <b>2.14</b> | <b>2.62</b> | <b>3.20</b> | <b>3.92</b> | <b>22.2%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 65** LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

| Country/Region        | 2021        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | 2028        | CAGR<br>(2023–2028) |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Brazil                | 0.10        | 0.12        | 0.14        | 0.16        | 0.19        | 0.22        | 0.26        | 0.30        | 16.5%               |
| Rest of Latin America | 0.12        | 0.14        | 0.16        | 0.19        | 0.22        | 0.25        | 0.29        | 0.34        | 16.6%               |
| <b>Total</b>          | <b>0.22</b> | <b>0.26</b> | <b>0.30</b> | <b>0.35</b> | <b>0.41</b> | <b>0.47</b> | <b>0.55</b> | <b>0.64</b> | <b>16.5%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis



## 8.5 OTHER TECHNOLOGIES

Thermo Fisher's ion semiconductor sequencing technology-based platforms include Ion Torrent PGM, Ion Proton, and the newly launched GeneStudio S5 Series. Ion semiconductor sequencing involves the direct translation of genetic codes of sequence (DNA bases) into digital information onto a semiconductor chip. The technology translates chemically encoded information (A, C, G, T) into digital information (0, 1) on a semiconductor chip. The company focuses on continuous enhancements in the technology to widen its scope of applications and increase its precision of sequencing.

The launch of new and advanced sequencers by Thermo Fisher and the advantages offered by ion semiconductor technology platforms are expected to drive market growth in the near future. For instance, in March 2022, the company launched the Ion Torrent Genexus Dx System. This fully validated system enables users to perform both diagnostic testing and clinical research on a single instrument. The company also provides complete NGS workflow for diagnostic solutions for comprehensive genomic profiling. The high adoption of ion semiconductor technology can be attributed to its features, including simpler workflow, time- as well as cost-efficiency, and scalability. This technology is gaining popularity in microbial profiling, thus contributing to the overall segment growth.

**TABLE 66** MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY REGION, 2021–2028 (USD MILLION)

| Region                 | 2021         | 2022         | 2023          | 2024          | 2025          | 2026          | 2027          | 2028          | CAGR<br>(2023–2028) |
|------------------------|--------------|--------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------------|
| North America          | 46.18        | 52.10        | 58.75         | 66.31         | 74.91         | 84.91         | 96.58         | 110.34        | 13.4%               |
| Europe                 | 25.14        | 28.51        | 32.32         | 36.67         | 41.63         | 47.44         | 54.24         | 62.28         | 14.0%               |
| Asia Pacific           | 6.96         | 8.08         | 9.36          | 10.85         | 12.58         | 14.62         | 17.04         | 19.94         | 16.3%               |
| Latin America          | 1.59         | 1.75         | 1.91          | 2.09          | 2.29          | 2.52          | 2.77          | 3.07          | 9.9%                |
| Middle East and Africa | 1.10         | 1.19         | 1.29          | 1.39          | 1.50          | 1.63          | 1.76          | 1.91          | 8.1%                |
| <b>Total</b>           | <b>80.97</b> | <b>91.63</b> | <b>103.64</b> | <b>117.32</b> | <b>132.92</b> | <b>151.11</b> | <b>172.40</b> | <b>197.53</b> | <b>13.8%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

**TABLE 67** NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

| Country      | 2021         | 2022         | 2023         | 2024         | 2025         | 2026         | 2027         | 2028          | CAGR<br>(2023–2028) |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|---------------------|
| US           | 41.85        | 47.25        | 53.33        | 60.24        | 68.11        | 77.27        | 87.96        | 100.57        | 13.5%               |
| Canada       | 4.33         | 4.85         | 5.42         | 6.07         | 6.80         | 7.65         | 8.62         | 9.77          | 12.5%               |
| <b>Total</b> | <b>46.18</b> | <b>52.10</b> | <b>58.75</b> | <b>66.31</b> | <b>74.91</b> | <b>84.91</b> | <b>96.58</b> | <b>110.34</b> | <b>13.4%</b>        |

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis